

Acute Effects of Capsaicin on Energy Expenditure and Fat Oxidation in Negative Energy Balance

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Background Addition of capsaicin (CAPS) to the diet has been shown to increase energy expenditure; therefore capsaicin is an interesting target for anti-obesity therapy. **Aim** We investigated the 24 h effects of CAPS on energy expenditure, substrate oxidation and blood pressure during 25% negative energy balance. **Methods** Subjects underwent four 36 h sessions in a respiration chamber for measurements of energy expenditure, substrate oxidation and blood pressure. They received 100% or 75% of their daily energy requirements in the conditions '100%CAPS', '100%Control', '75%CAPS' and '75%Control'. CAPS was given at a dose of 2.56 mg (1.03 g of red chili pepper, 39,050 Scoville heat units (SHU)) with every meal. **Results** An induced negative energy balance of 25% was effectively a 20.5% negative energy balance due to adapting mechanisms. Diet-induced thermogenesis (DIT) and resting energy expenditure (REE) at 75%CAPS did not differ from DIT and REE at 100%Control, while at 75%Control these tended to be or were lower than at 100%Control ($p = 0.05$ and $p = 0.02$ respectively). Sleeping metabolic rate (SMR) at 75%CAPS did not differ from SMR at 100%CAPS, while SMR at 75%Control was lower than at 100%CAPS ($p = 0.04$). Fat oxidation at 75%CAPS was higher than at 100%Control ($p = 0.03$), while with 75%Control it did not differ from 100%Control. Respiratory quotient (RQ) was more decreased at 75%CAPS ($p = 0.04$) than at 75%Control ($p = 0.05$) when compared with 100%Control. Blood pressure did not differ between the four conditions. **Conclusion** In an effectively 20.5% negative energy balance, consumption of 2.56 mg capsaicin per meal supports negative energy balance by counteracting the unfavorable negative energy balance effect of decrease in components of energy expenditure. Moreover, consumption of 2.56 mg capsaicin per meal promotes fat oxidation in negative energy balance and does not increase blood pressure significantly. Trial Registration Netherlands Trial Register; registration number NTR2944